

BONDING AND STRUCTURE

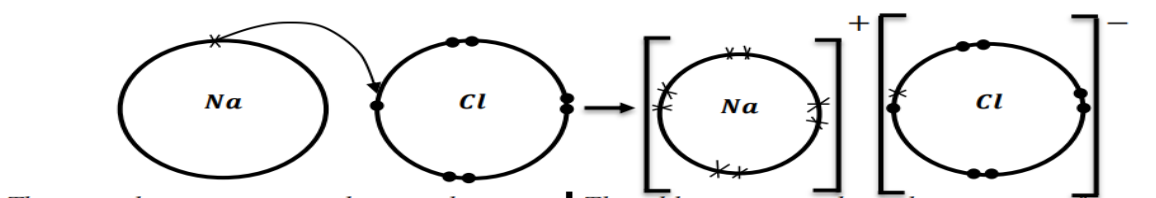
A **Chemical bond** is a force of attraction that holds atoms, ions or molecules together to make compounds. During formation of a chemical bond, atoms combine in order to acquire a noble gas electronic structure by either transfer of electrons from one atom to another or by sharing electrons contributed by each atom or by electrostatic attraction between delocalized electrons and positively charged ions. This determines the type of bond to be formed.

IONIC BOND/ ELECTROVALENT BOND

It is formed by complete transfer of one or more electrons from one atom to another to form oppositely charged ions. The electron(s) are transferred from a metallic element to a non-metallic element. The element that loses electrons, the metal forms a positively charged ion (cation) and so is more **electropositive** while the element gaining electrons is **electronegative** and forms a negatively charged ion (anion). The cation and anion formed are held together by strong forces of attraction, the **electrostatic forces of attraction**.

Let us consider the formation of sodium chloride;

- ✓ The sodium atom has electronic configuration, $1s^2 2s^2 2p^6 3s^1$. The outermost energy level has only one electron and thus unstable. This electron is transferred to the chlorine atom by sodium to form noble gas, stable configuration; $1s^2 2s^2 2p^6$ and hence sodium ion is formed.
- ✓ The chlorine atom on the other hand has electronic configuration, $1s^2 2s^2 2p^6 3s^2 3p^5$. The outer most energy level has seven electrons and thus unstable. This only lacks one electron to form a noble gas, stable configuration; $1s^2 2s^2 2p^6 3s^2 3p^6$. This gains an electron from the sodium atom and hence chloride ion is formed the sodium ion and chloride ion get hold together by strong electrostatic forces of attraction as illustrated below;



Mind problem;

With the aid of an illustration, describe the formation of;

- a) *Magnesium chloride* b) *Potassium oxide by electrovalent bonding.*

GIANT IONIC STRUCTURES

In giant ionic structures, metallic elements combine with non-metallic elements forming strong electrostatic attractions in the compounds between the ions of opposite charges, the cations and anions. Each ion is surrounded by the greatest number of oppositely charged ions called the coordination number.

Examples of compounds with giant ionic structures include the following;

<i>Compound</i>	<i>Structure</i>
Sodium oxide	$(Na^+)_2O^{2-}$
Magnesium oxide	$Mg^{2+}O^{2-}$
Calcium chloride	$Ca^{2+}(Cl^-)_2$
Sodium hydride	Na^+H^-

Properties of ionic compounds

High melting and boiling points: Ionic compounds have strong electrostatic forces of attraction between the oppositely charged ions that require a high amount of energy to break.

They conduct electricity when in molten or aqueous state: In the ionizing solvent, the forces of attraction between the ions are greatly reduced and the oppositely charged ions become free to move in molten or aqueous state under the influence of an electric field.

They readily dissolve in polar solvents such as water but insoluble in non-polar solvents such as methylbenzene The ion-solvent interactions are stronger than ion-ion interactions of the compound for polar solvents hence soluble, Ion-ion interaction is stronger than ion-solvent interaction and solvent-solvent interaction for non-polar solvents making the compound insoluble.

- ✓ On dissolving in water, the electrostatic forces of attraction between the cations and anions in the crystal lattice are greatly reduced. The ions become free to move in solution. Since water is a polar molecule, the cations get attracted to the negative end of water molecules and the anions get attracted to the positive end of the water molecules. The formation of ion-solvent interactions releases enough energy to break the electrostatic forces of attraction. The ion-solvent interactions are stronger than ion-ion interactions in the compound hence soluble in water. On dissolving in a non-polar solvent, the Ion-ion interaction is stronger than ion-solvent interaction and solvent-solvent interaction in the non-polar solvent.

They exist as crystalline solids at room temperature: The strong electrostatic forces of attraction between the oppositely charged ions require a high amount of energy to break at room temperature, keeping the molecules so closely packed together.

Mind problems;

Explain why table salt conducts electricity when dissolved in water while sugar does not.

Explain what happens when table salt is dissolved in water.

Why is table salt preferred to sugar for preserving meat?

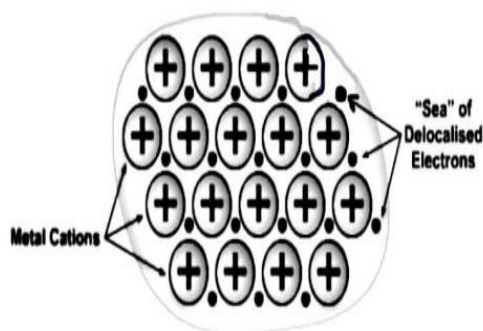
In Uganda, aluminium is extracted from bauxite ore and during the concentration of the ore, aluminium oxide is obtained. This process is essential in the manufacturing of materials such as roofing sheets, cooking utensils and electric cables. A group of students visited one of the bauxite rich areas in Kasese to study some properties of the purified ore, Al_2O_3 and make the following observations during their study. Aluminium oxide has a very high melting point, does not conduct electricity in solid form but does so in molten state and is stronger than sodium chloride on comparison of ionic strength;

- a) Explain why aluminium oxide has a very high melting point which is even higher than that of sodium chloride.*
- b) Why doesn't solid aluminium oxide conduct electricity but does so in molten state?*
- c) The students observed that both sodium chloride and aluminium oxide dissolve in water but sodium chloride dissolves more readily than aluminium oxide. Explain this observation.*

METALLIC BONDING

A metallic bond is the electrostatic force of attraction between positive metal ions and delocalized electrons.

A metallic solid consists of a lattice of metal ions with the outer most electrons forming a "sea" of delocalized electrons which freely move throughout the structure and can conduct electricity. Electrostatic attractions exist between the delocalized electrons and the metal ions.



Properties of metals

All the properties of metals are as a result of metallic bonding that occurs within metals and these include;

High melting and boiling points; Metallic bonds are very strong and require high amounts of energy to break. The strength of the metallic bond increases as the number of electrons contributed per atom of a metal towards metallic bonding increases hence increasing the melting and boiling points.

- ✓ This explains why group II elements form metals with higher melting and boiling points than group I elements; Both group I and group II elements form metallic bonds whose strength depend on the number of electrons contributed per atom to the formation of the metallic bonds. Group II elements contribute two electrons per metal atom towards metallic bond formation forming stronger metallic bonds that require a higher amount of energy to break than group I elements which only contribute one electron per metal atom towards formation of metallic bonds forming weaker metallic bonds that require a lower amount of energy to break hence lower melting points.
- ✓ Transition elements have higher melting points than the main group elements; this is because transition elements have smaller atomic radii and contribute both the 4s electron and the 3d electrons forming stronger metallic bonds that require a higher amount of energy to break compared to the main group elements.

Mind probe;

- 1) Explain why chromium has a higher melting point than that of calcium.
- 2) Compare the plausible melting points of element X (atomic number 19) and element Y (atomic number 20) and explain the differences.

- 3) The melting points of Sodium, Magnesium and aluminium are 97.8, 650 and 660.3°C respectively. Explain the general observable trend in the melting points of the elements.

Electrical conductivity; Metals are good conductors of electricity and heat due to presence of free mobile electrons throughout the lattice. If a potential difference is applied between the ends of a metal, the delocalized electron cloud will move towards the positive potential. However, electrical conductivity of a metal decreases with increase in temperature because resistance increases. This is because increase in temperature increases thermal vibration within the metal. The regularity within the metallic lattice is upset and therefore the electrons cannot easily move in the lattice.

Thermal conductivity; Metals conduct heat due to exchange of kinetic energy between free mobile electrons in the lattice. When heat is supplied to one end of a piece of a metal, the kinetic energy of the electrons is increased. This increase in kinetic energy is transmitted through the system of delocalized electrons to other parts of the metal.

Malleability and ductility; Metals are malleable (can be hammered-beaten into sheets) and ductile (can be drawn out under tension-pulled out into wires). This is due to the ability of layers of atoms to roll over each other into new positions without breaking the metallic bond.

Insoluble in polar and non-polar solvents; the metal-metal attractions are much stronger than the solvent-solvent attractions or the metal-solvent attractions. The solvent molecules cannot penetrate the metal lattice.

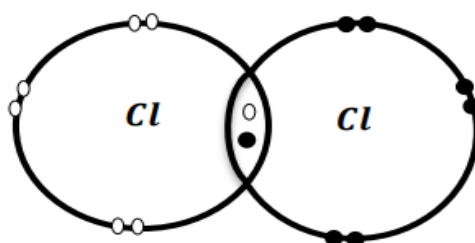
COVALENT BONDING

A covalent bond is a bond formed when electrons are shared in pairs between two bonded atoms.

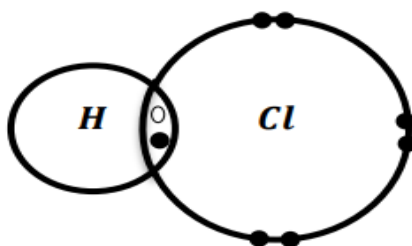
In a single covalent bond between two atoms, one electron is contributed by each atom and is shared as a pair of electrons. The shared pair of electrons is called a covalent bond. The shared pair of electrons occupies the same orbital with opposing spins and each of the electrons must be unpaired from the orbital in which it is got. In double covalent bonds, two pairs of electrons are shared and in a triple covalent bond, three pairs of electrons are shared.

Covalent bonds may be formed between atoms of the same element or atoms of different elements. Usually, the bonding atoms contribute an equal number of electrons towards formation of a normal covalent bond.

Let us consider the formation of chlorine molecule (Cl_2) between chlorine atoms; The chlorine atom has electronic configuration, $1s^2 2s^2 2p^6 3s^2 3p^5$. The chlorine molecule is formed by each of the chlorine atom sharing one of its valence electrons with another atom. The two atoms approach each other closely for their atomic orbitals to overlap as illustrated below;



Considering the formation of Hydrogen chloride (HCl), the hydrogen atom has electronic configuration $1s^1$ and the chlorine atom has electronic configuration, $1s^2 2s^2 2p^6 3s^2 3p^5$. Hydrogen contributes its only electron and chlorine contributes its unshared valence electron. The electron pair is shared to make hydrogen have a full sub energy level and chlorine to have a full octet as shown below;



Mind probe;

Illustrate the formation of Carbon dioxide, Ethene, Water and Ammonia by covalent bonding;

DATIVE COVALENT BONDING

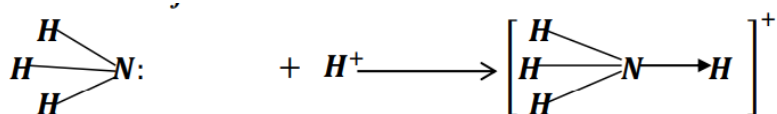
A dative covalent bond or coordinate covalent bond is a covalent bond formed when the shared pair of electrons is contributed by only one of the bonded atoms.

The atom that contributes the pair of the electrons is the donor atom and the other atom is the acceptor atom. For an atom to act as a donor atom, it must have at least one pair of unshared electrons, called the lone pair. For an atom to act as an acceptor atom, it must have a vacant orbital(s). An acceptor can be a transition metal atom or an atom in a molecule. An arrow is always drawn from the donor atom to the acceptor atom to illustrate electron movement and hence a dative bond.

Examples of dative bonding in compounds include;

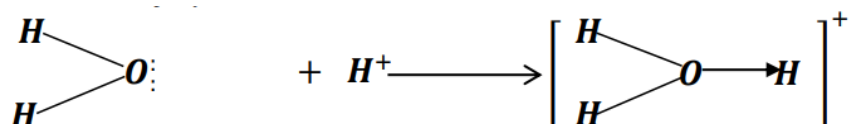
a) Formation of ammonium ion;

This ion is formed by coordination of ammonia to a hydrogen ion. The nitrogen atom in the ammonia molecule has a lone pair of electrons. The hydrogen ion has electronic configuration $1s^0$. The hydrogen ion therefore accepts the lone pair of electrons from nitrogen to have a full $1s$ sub energy level. This results into formation of the ammonium ion.

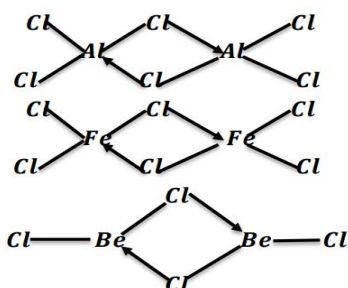


b) Formation of hydroxonium ion;

This ion is formed by coordination of water to a hydrogen ion. The oxygen atom in the water molecule has two lone pairs of electrons. The hydrogen ion has electronic configuration $1s^0$. The hydrogen ion therefore accepts one of the lone pairs of electrons from oxygen to have a full $1s^2$ sub energy level. This results into formation of the hydroxonium ion as shown below;

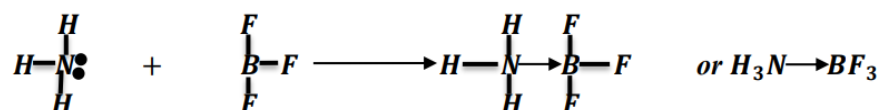


c) Formation of dimerised aluminium chloride , dimerised iron(III) chloride and dimerised beryllium chloride;



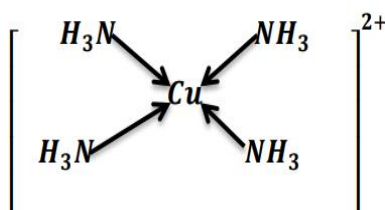
d) Reaction of Ammonia and boron trifluoride;

The nitrogen atom in the ammonia molecule has a lone pair of electrons. The boron atom has only three valence electrons. When boron covalently bonds with three fluorine atoms in boron trifluoride, it has six electrons in its outermost energy level, requiring two electrons to complete its octet. Since nitrogen is more electronegative than hydrogen in ammonia, it has a partial negative charge. Also boron is less electronegative than fluorine in boron trifluoride, hence has a partial positive charge. There is therefore an attraction between nitrogen atom and boron atom. The lone pair of electrons on the nitrogen atom is donated to the boron atom to form a dative bond. This also occurs between ammonia and aluminium chloride, boron trichloride.



e) Formation of transition metal complexes;

The donor atoms in ligands donate lone pairs of electrons to the central metal atoms. These electrons occupy the empty d-orbitals of the transition metal atoms forming an ion, the complex ion. An example is the formation of tetraamminecopper (II) ion.



Properties of covalent compounds

Low melting and boiling points; there are weak forces of attraction between the molecules which require low amounts of energy to break.

They do not conduct electricity except graphite; they neither form free mobile ions nor have delocalized electrons to conduct electricity.

Most covalent substances are insoluble in polar solvents like water but soluble in non-polar solvents; in polar solvents, the solvent-solvent interaction is much greater than interaction between the covalent molecules in the solute or the interaction between the covalent molecules and the polar molecules in the solution. Covalent compounds tend to dissolve in non-polar solvents because van der Waals' forces are the binding forces in both cases. However, some covalent compounds containing polar bonds as or may dissolve in water due to interaction between the polar bond and the water.

Exist as gases or liquids at room temperature; there are weak van der Waals' forces of attraction between the molecules that require a low amount of energy to break at room temperature, keeping the molecules far apart at room temperature.

Covalent character in ionic compounds

Some ionic compounds tend to behave as covalent compounds; this is explained in terms of polarising power and polarisability.

1. Polarising power;

This is the power of a cation to distort the electron cloud of an anion in contact with it. The greater the polarising power of the cation, the greater the tendency of the cation to form a covalent bond with the anion, electrons are partly shared between the cation and anion hence covalency.

Polarising power of a cation depends on;

- (i) **Charge of the cation;** Cations with high charges have high polarising power. The larger the positive charge on the ion, the greater the attraction of the valence electrons. The ion therefore has a higher polarising power than the due to its higher charge.
- (ii) **Cationic radius;** the smaller the ionic radius, the higher the polarising power of the cation. The ion therefore has a higher polarising power than the due to its smaller ionic radius.
- (iii) **Charge density of the cation;** Charge density is the ratio of ionic charge to ionic radius. The higher the charge density, the higher the polarising power of the cation.

2. Polarisability;

This is the ease with which the electron cloud of the anion can be distorted by the cation in contact with it. The greater the polarisability of the anion, ease its electron cloud can be distorted and hence the greater the tendency of the anion to form a covalent bond.

Polarisability of an anion depends on the radius of the anion; the smaller the anion, the lower it is polarisable and the larger the anion, the easier it is polarized.

Mind probe;

1. Explain why;

- (a) Calcium iodide is ionic whereas aluminium iodide is covalent.
- (b) Sodium chloride melts at 800°C whereas aluminium chloride sublimes at 180°C.
- (c) Beryllium chloride is more soluble in ethanol than in water whereas barium chloride is more soluble in water than in ethanol.
- (d) The solubility of lithium chloride in ethanol is higher than the solubility of potassium chloride in ethanol.

(e) Aluminium chloride is soluble in methylbenzene and has a low melting point yet aluminium fluoride is soluble in water and has a higher melting point.

2. The table below shows the melting points of the chlorides of Group II elements.

Chloride	$BeCl_2$	$MgCl_2$	$CaCl_2$	$SrCl_2$	$BaCl_2$
Melting point ($^{\circ}C$)	405	714	782	875	962

Explain why;

- (i) the melting point of beryllium chloride is very low
- (ii) melting point increases from magnesium chloride to Barium chloride

VALENCE SHELL ELECTRON PAIR REPULSION (VSEPR) THEORY AND THE SHAPES OF MOLECULES AND IONS

This theory states that electron pairs; the bonding pairs and lone pairs in the valence shell of a central atom repel each other and therefore arrange themselves as far apart as possible to achieve minimum repulsion and maximum stability. This arrangement determines the shape (geometry) of a molecule or ion. Bonding pairs are shared pairs of electrons that participate in bonding while lone pairs are pairs of unshared electrons.

Between one electron pair and any other electron pair, there is a force of electrostatic repulsion. Any lone pairs of electrons on the central atom repel the bonding pairs of electrons and this affects the geometry (bond angle and shape) of the molecule or ion.

For the purpose of determining electron geometry and shape, the VSEPR theory considers the electron pairs in a double bond or a triple bond are treated as one pair. Since the lone pairs are closer to the central atom, they cause greater repulsion than bonding pairs and the repulsions are in the order;

Lone pair-Lone pair repulsion > Lone pair-Bond pair repulsion > Bond pair-Bond pair Repulsion.

Steps followed when drawing the structures of molecules and cations

1. Determine the central atom; this is the least electronegative atom or the atom that occurs in the lowest ratio.

2. Determine the number of valence electrons on the central atom from its electronic configuration or the number of its group in the Periodic Table. If the structure is for a cation, subtract an electron for each positive charge on the total number of valence electrons and if it is for an anion, add an electron for each negative charge on the total number of valence electrons. If the central atom is a transition element, then calculate the oxidation state and use it as the total number of valence electrons.
3. Add one electron to the total valence electrons of the central atom obtained above for each atom/ ligand bonded to the central atom, depending on its charge as an ion.
4. Divide the total number of valence electrons by two to get the total number of electron pairs. This is because electrons bond in pairs
5. From the total number of electron pairs, subtract the number of bonding pairs to get the number of lone pairs. The number of bonding pairs corresponds to the number of bonded atoms to the central atom.
6. Then draw the structure of the molecule or ion.

Oxy-anions and oxy-molecules

These are molecules or anions containing oxygen atoms. They are treated in a special way since they contain **–OH** bonds, **=O** bonds and **–O[–]** bonds.

When drawing the shape of an oxy-anion or molecule the same rules of drawing structures of molecules are applied;

- Deduce the nature of bonds in the ion or molecule. Begin with **–O[–]** bonds followed by **–OH** bonds and then **=O** bonds.
 - (i) For every negative charge in the anion, there is one **–O[–]** bond.
 - (ii) For every hydrogen atom, there is one **–OH** bond.
 - (iii) All the remaining oxygen atoms are double bonded after (i) and (ii) or if (i) and (ii) do not apply to the molecule.
- Since for each single bond, one electron of the valence electrons of the central atom is used, and two for a double bond, the unbonded electrons are the lone pairs.
- Then draw the structure of the molecule/ion using the nature of bonds and lone pairs.

SHAPES ATTAINED BY MOLECULES AND IONS

The shape of the molecule or ion is deduced by the nature of bonds and lone pairs. **The number of bonding pairs and lone pairs** therefore guide us on the shape of the molecule. Bond angles and bond lengths should be put into consideration when drawing the structures and the lines that represent the bonds should not be hanging, do not leave a lot of space between the atoms.

a) Linear Shape

Molecules have two bonding pairs and no lone pairs. The molecules are linear with a bond angle of 180° because the bonding pairs greatly repel themselves.

Let us consider the shape of Carbon dioxide gas; CO_2

- ✓ The carbon atom is the central atom and has 4 valence electrons from its electronic configuration. Carbon is therefore expected to form four bonds with the oxygen atoms. Each oxygen atom forms a double bond to the carbon atom and no lone pair remains. The two bonding pairs greatly repel themselves greatly to form a linear molecule with the bond angle of 180° .

Other molecules include; Beryllium chloride, carbon disulphide, ethyne, Hydrogen cyanide, among others.

<i>Molecule</i>	<i>Structure</i>
<i>BeCl_2</i>	<i>$\text{Cl} - \text{Be} - \text{Cl}$</i>
<i>CO_2</i>	<i>$\text{O} = \text{C} = \text{O}$</i>

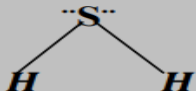
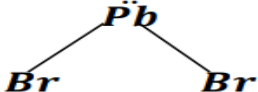
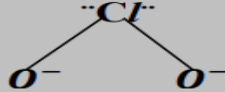
b) Bent shape (V-shaped molecules)

Molecules have two bonding pairs and one lone pair or two bonding pairs and two lone pairs. The molecules are V- shaped with a bond angle of about 92° due to the repulsion between the two lone pairs of electrons which is greater than the repulsion between lone pairs and bonding pairs of electrons which is also greater the repulsion between the bonding pairs of electrons reducing the bond angle greatly.

Let us consider the shape of Hydrogen sulphide; H_2S

- ✓ Sulphur atom is the central atom and has 6 valence electrons from its electronic configuration. Since it is bonded to two hydrogen atoms, the total number of valence electrons in the molecule is 8. This gives four electron pairs. Of the four electron pairs, there are two bonding pairs since Sulphur is bonded to two hydrogen atoms and two lone pairs. The repulsion between the two lone pairs of electrons is greater than the repulsion between the lone pairs and bonding pairs of electrons which is also greater the repulsion between the bonding pairs of electrons. This reduces the bond angle making the molecule V-shaped.

Other molecules include Water, tin (ii) chloride, Lead (ii) bromide, Chlorate(iii), among others.

<i>Molecule</i>	<i>Structure</i>
H_2S	
$PbBr_2$	
ClO_2^-	

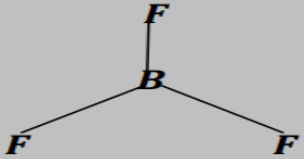
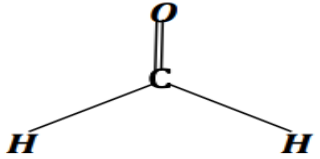
c) **Trigonal Planar shape;**

Molecules have three bonding pairs and no lone pair. The three bonding pairs lie in the same plane and the molecules have a bond angle of 120° because the bonding pairs greatly repel each other equally.

Let us consider the structure of Boron trifluoride; BF_3

- ✓ The boron atom is the central atom and has 3 valence electrons from its electronic configuration. Since it is bonded to three fluorine atoms, the total number of valence electrons in the molecule is 6. This gives three electron pairs. All the three electron pairs are used in bonding to form three bonding pairs since boron is bonded to three fluorine atoms and therefore there is no lone pair. The three bonding pairs of electrons repel each other equally. This makes the bond angle 120° making the molecule trigonal planar.

Other molecules include; methanal, Boron trichloride, Nitrate ion, carbonate ion, hydrogen carbonate ion, among others.

<i>Molecule</i>	<i>Structure</i>
BF_3	
$HCHO$	

d) **Tetrahedral shape;**

Molecules or ions have four bonding pairs of electrons and no lone pairs. They have a bond angle of approximately 109.5° because the bonding experience mutual repulsion.

Let us consider the shape of silicon tetrachloride;

- ✓ The silicon atom is the central atom and has 4 valence electrons from its electronic configuration. Since it is bonded to four chlorine atoms, the total

number of valence electrons in the molecule is 8. This gives four electron pairs. All the four electron pairs are bonding pairs since silicon is bonded to four chlorine atoms and no lone pairs. The four bonding pairs of electrons mutually repel each other, resulting into a bond angle of 109.5° making the molecule tetrahedral.

Other molecules include methane, ammonium ion, carbon tetrachloride, Sulphuric acid, Phosphoric acid, sulphate ion, chromate (vi) ion, among others.

<i>Molecule</i>	<i>Structure</i>
$SiCl_4$	
PO_4^{3-}	

e) Trigonal Pyramidal shape;

Trigonal pyramidal molecules or ions have three bonding pairs of electrons and one lone pair. They have a bond angle smaller than which is observed in the tetrahedral molecules. This is due to increased repulsion caused by the lone pair.

Let us consider the structure of phosphine; PH_3

- ✓ The phosphorus atom has 5 valence electrons from its electronic configuration. Since it is bonded to three hydrogen atoms, the total number of valence electrons in the molecule is 8. This gives four electron pairs. Of the four electron pairs, there are three bonding pairs since phosphorus is bonded to three hydrogen atoms and one lone pair. The three bonding pairs of electrons repel each other but the lone pair-bond pair repulsion is greater. This reduces the bond angle to 107° making the molecule trigonal pyramidal.

Other molecules include; ammonia, chlorate (vi) acid, sulphite ion, phosphoric acid, hydroxonium ion, phosphorous trichloride, phosphorous tri bromide, among others.

<i>Molecule</i>	<i>Structure</i>
PH_3	
$HClO_3$	
SO_3^{2-}	

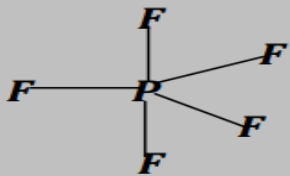
f) Trigonal bipyramidal shape;

Molecules have five bonding pairs of electrons and no lone pairs.

Let us consider Phosphorous pentafluoride; PF_5

- ✓ The phosphorus atom is the central atom and has 5 valence electrons from its electronic configuration. Since it is bonded to five fluorine atoms, the total number of valence electrons in the molecule is 10. This gives five electron pairs. All the five electron pairs are used in bonding as bonding pairs and thus no lone pair.

Other molecules include Phosphorous pentachloride, Phosphorous pentaiodide, among others.

<i>Molecule</i>	<i>Structure</i>
PF_5	

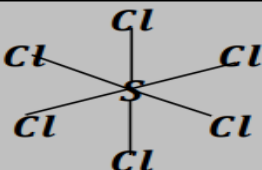
g) Octahedral shape;

Molecules have six bonding pairs of electrons and no lone pair.

Let us consider Sulphur hexachloride;

- ✓ The sulphur atom is the central atom and has 6 valence electrons from its electronic configuration. Since it is bonded to six chlorine atoms, the total number of valence electrons in the molecule is 12. This gives six electron pairs. All the six electron pairs are used in bonding as bonding pairs.

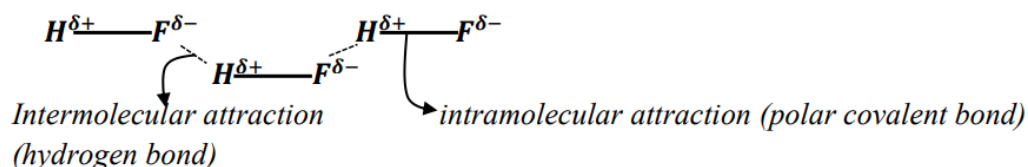
Other molecules include sulphur hexabromide, sulphur hexafluoride, among others.

<i>Molecule</i>	<i>Structure</i>
SCl_6	

INTERMOLECULAR FORCES AND POLARITY OF MOLECULES

These are forces that exist between molecules. They include **Van der Waals forces** and the **hydrogen bond**. They are different from forces holding individual atoms in a molecule. The forces between atoms in a molecule are called the intramolecular forces.

Intermolecular forces	Intramolecular forces
van der Waals forces	Covalent bond
Hydrogen bonds	Ionic bond
	Metallic bond



VAN DER WAALS FORCES

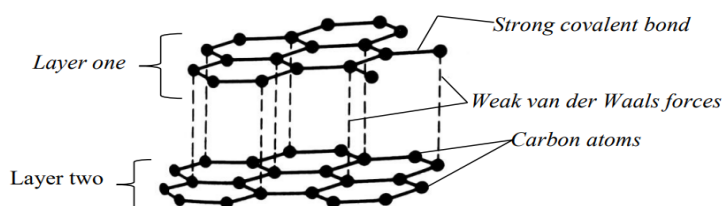
These forces are neither ionic nor covalent. The Van Der Waals forces may exist between polar molecules and these are called **dipole-dipole interactions**. The van der Waals forces that exist between non-polar molecules are called **London dispersion forces** (temporary dipole-dipole attractions).

Dipole- dipole interactions exist between polar molecules like hydrogen chloride, ammonia among others. These molecules interact to form an ordered arrangement. In this arrangement, partial positive charge in one molecule is adjacent to partial negative charge in another molecule. The dipole-dipole attractive forces are increased by one dipole inducing a dipole in another neighboring molecule.

Van der Waals forces are in general very weak compared to ionic and covalent bonds. This is shown by the low melting and boiling points of substances held together by van der Waals" forces.

Explanations based on Van der Waals forces;

1) Graphite is soft and slippery;



- ✓ Each carbon atom in graphite is covalently bonded to three other carbon atoms to form two dimensional layers of hexagonal rings of carbon atoms. The hexagonal layers held together by weak Van der Waals forces of attraction which allow the layers to slide over each other.

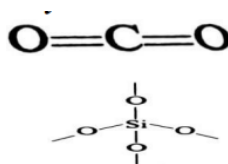
2) Variation in the melting points of period 3 elements;

Element	<i>P</i>	<i>S</i>	<i>Cl</i>	<i>Ar</i>
Melting point(⁰ C)	44.2	115.2	-101.5	-189.4

- ✓ Phosphorus, Sulphur, chlorine and argon have simple molecular structures with weak Van der Waals forces of attraction whose magnitude increases in strength with increase in molecular weight. Sulphur is octatomic (S_8) with a higher molecular weight hence stronger Van der Waals forces of attraction than Phosphorus which is tetratomic (P_4). Chlorine is diatomic (Cl_2), with a higher molecular weight and stronger Van der Waals forces than argon which is monoatomic. The energy required to break the Van der Waals forces therefore decreases from phosphorus to argon.

3) Carbon dioxide is a gas whereas silicon(IV) oxide is a solid at room temperature;

- ✓ Carbon dioxide is non-polar. It has discrete molecules with a simple molecular structure held by weak van der Waals forces of attraction which are easily overcome at room temperature keeping the molecules far apart from each other. However, in silicon (iv) oxide, each silicon atom is covalently bonded to four oxygen atoms to form a tetrahedral structure which is continuous to form a giant molecular structure with very many strong covalent bonds that are not easily broken at room temperature, keeping its molecules very close to each other.



4) Variation in the boiling points of the hydrides of group IV elements;

Hydride	CH_4	SiH_4	GeH_4	SnH_4	PbH_4
Boiling point(⁰ C)	-161.5	-111.9	-88	-52	-13

- ✓ Boiling point increases from methane to plumbane. This is because molecules of the hydrides are held by weak Van der Waals forces of attraction whose strength increases with increase in molecular weight and molecular size.

5) Physical states of halogens;

- ✓ Fluorine and chlorine exist as gases whereas bromine and iodine exist as liquid and solid respectively at room temperature. From fluorine to iodine, atomic radius increases,

molecular size and molecular mass increases, hence magnitude of Van der Waals' forces of attraction holding the molecules increase in the order; . The Van der Waals' forces in bromine and iodine are strong enough to hold the molecules closer to each other existing as liquid and solid respectively. Fluorine and chlorine have weaker Van der Waals' forces due to their small size and low molecular masses. Their molecules are therefore far apart.

6) Boiling points of branched and unbranched isomers of alkanes;

- ✓ Branched isomers have a spherical shape which reduces its size and weaker Vander Waals forces of attraction exist between the molecules. Straight chain molecules give the molecule an extended structure resulting into relatively stronger Vander Waals forces of attraction requiring high temperatures to boil hence higher boiling points.

7) Variation in boiling points of alkylhalides

Element	CH_3F	CH_3Cl	CH_3Br	CH_3I
Boiling point($^{\circ}C$)	-78.4	-24.0	5.0	43.0

- ✓ Boiling point increases from fluoromethane to chloromethane to bromomethane to iodomethane. This is because the atomic radius of the halogen atoms increases in the order; $F < Cl < Br < I$, molecular size and molecular mass of the alkylhalides increases, hence magnitude of Van der Waals forces of attraction holding the molecules increases.

Element	CH_3Cl	CH_3CH_2Cl	$CH_3CH_2CH_2Cl$	$CH_3CH_2CH_2CH_2Cl$
Boiling point($^{\circ}C$)	-24.0	12.5	47.0	78.5

- ✓ Boiling point increases from chloromethane to chloroethane to chloropropane to chlorobutane. This is because molecules of the chloroalkanes are held by weak Van der Waals forces of attraction whose strength increases with increase in molecular weight and molecular size.

THE HYDROGEN BOND

A hydrogen bond is a **dipole- dipole attraction** which occurs between a hydrogen atom attached to a strongly electronegative atom and a strongly electronegative atom with a lone pair of electrons. The electronegative atoms are commonly fluorine, oxygen or nitrogen. Fluorine has the highest electronegativity. For this reason, hydrogen fluoride forms the strongest hydrogen bonds which persist even in vapour state. The greatest number of hydrogen bonds known are those which unite two oxygen atoms. The hydrogen bond is **weaker than a normal covalent bond**.

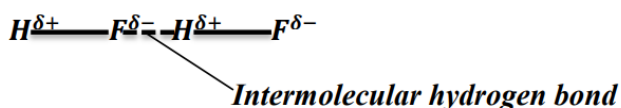
A hydrogen bond is either intermolecular if it occurs between two or more similar or different molecules and intramolecular if they exist within the same molecule.

Intermolecular hydrogen bonding occurs in the following;

Hydrogen fluoride, water and ammonia molecules;

Hydrogen fluoride;

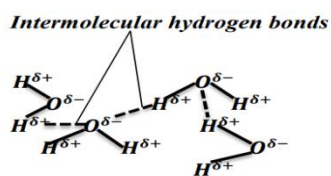
- If two hydrogen fluoride molecules are close enough, there is an attraction between the positive end of one molecule and the negative end of the other molecule. Since the attraction between the hydrogen atom in one molecule and the fluorine atom in the other molecule is stronger than the repulsion between the two hydrogen atoms and between the two fluorine atoms, the two molecules are bonded together by a hydrogen bond.



This explains why hydrogen fluoride is a **liquid** whereas other hydrides of group (vii) elements are gases at room temperature; hydrogen fluoride has a higher boiling point than any other hydride of any group (vii) element.

Water molecules;

- Water molecules associate through intermolecular hydrogen bonding. Each water molecule forms four hydrogen bonds with other water molecules. This explains why water has a higher boiling point and is a liquid whereas hydrogen sulphide has a lower melting point and is a gas although both sulphur and oxygen are group (vi) elements.



Ammonia;

- Both ammonia (NH₃) and phosphine (PH₃) are trigonal pyramidal but ammonia has a higher boiling point than phosphine. This is because nitrogen has a smaller atomic radius and is **more electronegative** than phosphorus. This makes the nitrogen-hydrogen bond strongly polar whereas the phosphorus-hydrogen bond is non-polar. Ammonia molecules are therefore held together by strong hydrogen bonds which require a high amount of energy to break than the weak Van der Waals forces in phosphine.

Mind probe;

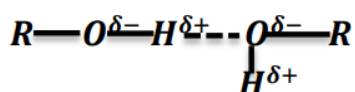
- 1) The boiling points of HF, H₂O and NH₃ are shown below. Explain the observed differences in boiling points.

<i>Molecule</i>	<i>Boiling point(°C)</i>
<i>HF</i>	19.9
<i>H₂O</i>	100
<i>NH₃</i>	-33.4

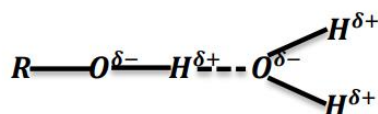
- 2) In Uganda, agriculture and clean energy development are crucial for sustainable development. Fertilizer industries produce chemicals like ammonia (NH₃), while clean energy projects encourage the use of clean gases like methane (CH₄) and carbon dioxide (CO₂). To ensure safe and efficient use of the substances, chemical engineers and technicians must understand their molecular structures and shapes. The ministry of science and technology is inviting secondary school learners to participate in a national science challenge.
- Draw the structures and name the molecular shapes of the compounds above using the VSEPR theory.
 - Explain briefly how the shape of ammonia affects its physical properties and uses in fertilizer production.
 - The bond angle in ammonium ion is 109° whereas that in ammonia is 107°. Explain.

Alcohols;

- Alcohols associate in their liquid form due to intermolecular hydrogen bonding. The boiling points of alcohols are higher than those of corresponding **alkanes** or **halogen compounds** with similar molar mass because of association through hydrogen bonding. For example the boiling point of methanol is whereas that of methane is



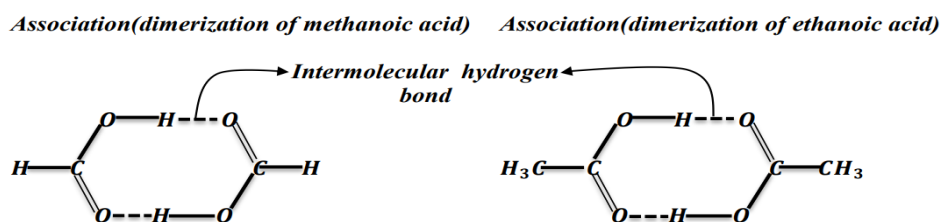
- Alcohols are also more soluble in water than alkanes or alkyl halides of similar molecular mass because of the ability to form hydrogen bonds with water.



- Alcohols with more hydroxyl groups have higher boiling points and are more soluble than alcohols with one hydroxyl group. Viscosity of alcohols also increases as number of hydroxyl groups increases due to increased hydrogen bonding.

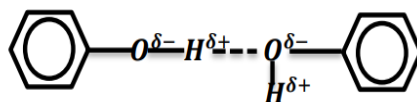
Carboxylic acids;

When Methanoic acid or ethanoic acid is dissolved in benzene or other organic solvents and the relative molecular mass of the acid determined by **cryoscopic method**, the value obtained is normally twice the actual value. This is because in liquid state form or in solution in benzene, the acids associate through intermolecular hydrogen bonds to form dimers. The same scenario happens when measurements in the vapour phase of the acids are done. In aqueous solution however, molecules of carboxylic acids do not dimerise but form intermolecular hydrogen bonds with water. This explains their solubility in water.



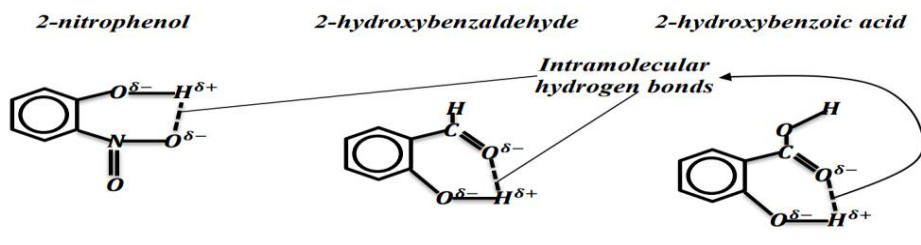
Phenols;

- Phenols associate in their liquid form due to intermolecular hydrogen bonding. The boiling points of phenols are higher than boiling points of corresponding alkanes or halogen compounds because of association through hydrogen bonding. Phenols are more soluble in water than corresponding alkanes or alkyl halides because of the ability to form hydrogen bonds with water just like alcohols.



Intramolecular hydrogen bonding

This is the type of hydrogen bonding that occurs within the same molecule. Intramolecular hydrogen bonding occurs in molecules such as 2-nitrophenol, 2-chlorophenol, 2-hydroxybenzaldehyde and 2-hydroxybenzoic acid



- The intramolecular hydrogen bonding prevents intermolecular hydrogen bonding between two or more molecules. Intramolecular hydrogen bonding is not possible for 3- and 4- substituted isomers (Para and Meta isomers) because of the substituent groups are far apart. However the Para and Meta isomers can form intermolecular hydrogen bonds. The intramolecular hydrogen bonding in ortho- substituted compounds explains their difference in physical properties from their Para- and meta-substituents. For example 2- nitrophenol has a low boiling point as compared to 3-nitrophenol and 4-nitrophenol.

The effect of hydrogen bonding;

- a) **Hydrogen fluoride is a liquid whereas other hydrides are gases at room temperature;**

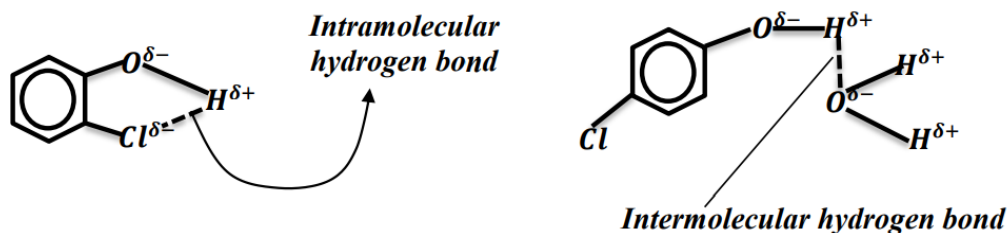
Fluorine atom has the smallest atomic radius and highest electronegativity among the halogens making the hydrogen-fluorine bond highly polar. The molecules of hydrogen fluoride are held by strong intermolecular hydrogen bonds which keep the molecules closer to each other. Molecules of hydrogen chloride, hydrogen bromide and hydrogen iodide are held by weak Van der Waals' forces that keep the molecules far apart.

- b) **Boiling points of hydrides of group(vii) elements;**

Hydride	<i>HF</i>	<i>HCl</i>	<i>HBr</i>	<i>HI</i>
Boiling point(^o C)	+19.9	-85.0	-66.7	-35.4

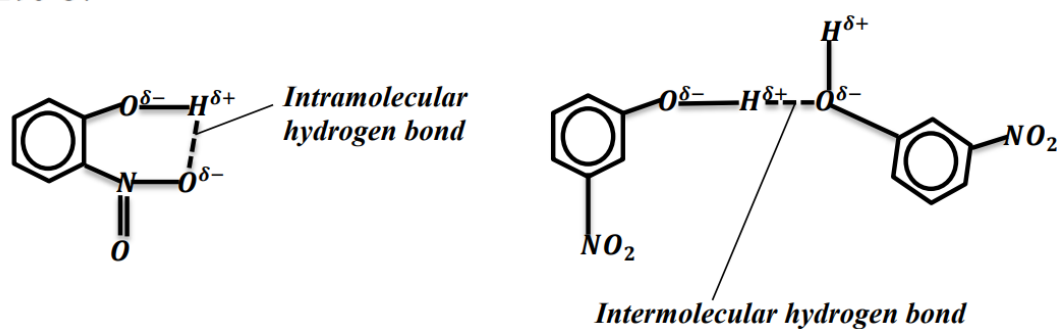
- Boiling point increases from hydrogen chloride to hydrogen iodide. However, hydrogen fluoride has an abnormally high value of boiling point. The increase in melting point from hydrogen chloride to hydrogen iodide is because the three hydrides have simple molecular structures held by weak Van der Waals forces whose magnitude increases with the increasing molecular weight leading to an increase in amount of energy required to break the forces. Hydrogen fluoride has an abnormally high melting point because the fluorine atom has the smallest atomic radius and the highest electronegativity, making the hydrogen-fluorine highly polar. Hydrogen fluoride molecules are therefore held by strong intermolecular hydrogen bonds which require a high amount of energy to break.

c) 2-chlorophenol is less soluble in water than 4-chlorophenol;



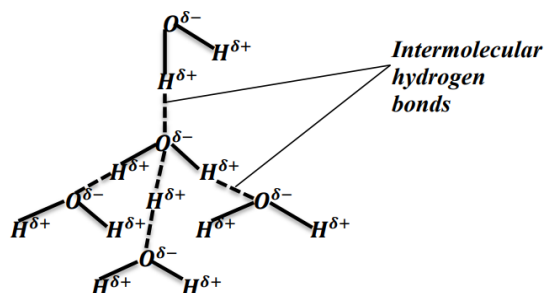
- Molecules of 2-chlorophenol form intramolecular hydrogen bonds because the chloro group is nearer the hydroxyl group. This suppresses the hydroxylic character of the compound or prevents formation of intermolecular hydrogen bonds between 2-chlorophenol and water molecules reducing its solubility. However, molecules of 4-chlorophenol cannot form intramolecular hydrogen bonds because the chloro group is far from the hydroxyl group. Molecules of 4-chlorophenol therefore form intermolecular hydrogen bonds with water. This increases its solubility.

d) The boiling point of 2-nitrophenol is 214°C whereas that of 3-nitrophenol is 290°C;



- Molecules of 2-nitrophenol form intramolecular hydrogen bonds because the nitro group is nearer the hydroxyl group. This suppresses the hydroxylic character of the compound or prevents formation of intermolecular hydrogen bonds between 2-nitrophenol molecules hence only weak van der Waals' forces hold the molecules. However, molecules of 3-nitrophenol cannot form intramolecular hydrogen bonds because the nitro group is far from the hydroxyl group. Molecules of 3-nitrophenol therefore form intermolecular hydrogen bonds which are stronger hence not easily broken.

e) Ice is less dense than water;



- In ice, each oxygen atom in a water molecule is surrounded tetrahedrally by four other oxygen atoms through extensive intermolecular hydrogen bonds. This gives ice an open structure. The water molecules in ice are spaced and further apart than they are in liquid water. Therefore ice occupies a bigger volume than that of liquid water of the same mass. Since density is inversely proportional to volume, ice has a lower density than water. However when ice melts, the hydrogen bonds break down and the molecules pack closely together so that liquid water has a lower volume hence a higher density than ice.

Mind probe;

Explain the following observations using the knowledge of hydrogen bonding and Van der Waals forces.

- 1) Hydrogen fluoride has a higher boiling point than hydrogen chloride.
- 2) Melting point of 4 – nitrophenol is much higher than that of 2 – nitrophenol.
- 3) 4-nitrophenol is more soluble in water than 2-nitrophenol.
- 4) 2-hydroxybenzaldehyde has a lower boiling point than 3-hydroxybenzaldehyde although the relative molecular masses of the two compounds are the same.
- 5) Boiling points of the given homologous series decreases in the order; Carboxylic acids > Alcohols > Amines > Alkanes of the same molecular mass.
- 6) Methyl amine (Molar mass = 31g) boils at -6.3°C while methanol (Molar mass = 32g) boils at 46°C.
- 7) Ethane boils at lower temperature than butane.
- 8) Water(Molar mass = 18g) is a liquid at room temperature whereas hydrogen sulphide (Molar mass = 34g) is a gas at room temperature

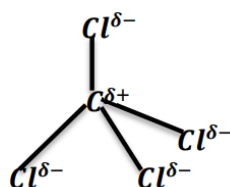
Polarity of molecules

Polarity is a condition of a molecule to have positive and negative charges. Polarity of molecules is related to the electronegativity difference of the bonding atoms and bond symmetry.

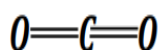
Polar covalent bonds are formed as a result of difference in electronegativity of the bonding atoms. These include; $H^{\delta+}-Cl^{\delta-}$, $H^{\delta+}-Br^{\delta-}$, $H^{\delta+}-I^{\delta-}$, $H^{\delta+}-O^{\delta-}$, $C^{\delta+}-Cl^{\delta-}$, etc.

For two equal but opposite separated electrical charges, there exists an electric dipole moment. An electric dipole moment is a product of one of the charges, or and the distance between the atomic nuclei. Any individual polar covalent bond must have a corresponding dipole moment. However, there are molecules in which the individual bonds in it are polar but the molecule as a whole is non-polar.

- ✓ An example is tetrachloromethane (CCl_4). Carbon tetrachloride is non-polar yet the carbon-chlorine bond is polar. Chlorine is more electronegative than carbon. The chlorine atom therefore tends to attract the bonding electrons towards itself hence acquires a partial negative charge (δ^-) and the carbon atom gains a partial positive charge (δ^+). This makes the carbon- chlorine bond polar. However, the whole molecule is non-polar because it has a tetrahedral structure in which the chlorine atoms are symmetrically distributed about the carbon atom. This creates equal and opposite dipole moments in all directions that cancel out making the resultant dipole moment zero.



- ✓ Another example is the carbon dioxide molecule (CO_2). The carbon- oxygen bond is polar but carbon dioxide is non-polar. Oxygen is more electronegative than carbon. The oxygen atom therefore tends to attract the bonding electrons towards itself hence acquires a partial negative charge (δ^-) and the carbon atom gains a partial positive charge (δ^+). This makes the carbon- oxygen bond polar. However, the whole molecule is non-polar because it has a linear structure in which the oxygen atoms are symmetrically distributed about the carbon atom. This creates equal and opposite dipole moments in all directions that cancel out making the resultant dipole moment zero.



- ✓ All molecules in which the atoms are symmetrically distributed around the central atom

are non polar. This is only possible if all the symmetrically distributed atoms around the central atom are the same. These include; linear molecules, trigonal planar molecules, tetrahedral molecules, trigonal bipyramidal molecules and octahedral molecules. Molecules in which the bonded atoms are not symmetrically distributed around the central atom like chloromethane, dichloromethane, trichloromethane, water and ammonia have an overall dipole moment over the molecule. Therefore both the individual bonds and the whole molecule are polar.

Mind probe;

Discuss why understanding the shape and polarity of water and carbon dioxide is important for environmental protection efforts in Uganda.

Differences between polar and non-polar molecules;

Polar molecules	Non-polar molecules
Asymmetrical	Symmetrical
One end of the molecule is positive whereas the other has a positive charge	Does not have distribution of charges at opposite ends
Can interact through hydrogen bonds	There are van der Waals forces between non polar molecules
Have a dipole moment	Have no dipole moment

Mind probe;

Explain why;

- 1) Trichloromethane is deflected by a charged rod but tetrachloromethane is not.
- 2) Boron trichloride is non-polar whereas nitrogen trichloride is polar.
- 3) A stream of chloroform is deflected by a charged rod but a stream of tetrachloromethane is not.

END, ENJOY YOUR READING.....